

# Ballistic mouse movements: EEG findings

Prepared for Prof. Serafeim Perdikis

Initial event audit and session-separated analysis of the available dataset.

## Scope and principal findings

A new MATLAB pipeline processed **380 run recordings from 25 participants**. Pre, post and retest were analysed separately. Event validation retained 35,210 trial sequences; subsequent quality rules retained **30,353 two-second epochs**. Twenty-three runs had no validated trials. S1 denotes spe30 and S2 denotes mle01.

**A confirmed label discrepancy was recovered.** In mle01 retest run 3 (13 November 2018), trials 76 and 78 contain displacement marker 500 and a 400-pixel leftward target jump in the mouse log, but their legacy labels are 1 (non-displaced). Their raw EEG segments match the saved epochs. The new metadata labels both as 2 (displaced); both are excluded from averaging because of data-loss overlap.

**The derived event parser missed nonzero-to-nonzero transitions.** A zero-to-nonzero rule reproduces all 398 legacy event-list entries for that run, explaining omission of 255-to-500 transitions. The original generating source was not found. Code 255 denotes data loss; the displacement markers themselves survive in the raw and saved trigger vectors.

**Recorded event timing warrants clarification.** The earlier S1/S2 EEG audit found median 400-to-500 marker intervals of 84 ms and 70 ms, respectively, exceeding the short paper's stated maximum 6 ms. Mouse logs corroborate nonzero intervals. These are software-marker intervals, not independent measurements of physical movement or screen onset.

## Accepted epochs contributing to figures

| Dataset          | Pre   | Post   | Retest |
|------------------|-------|--------|--------|
| S1 / spe30       | 496   | 498    | 393    |
| S2 / mle01       | 475   | 335    | 374    |
| All participants | 9,691 | 10,600 | 10,062 |

**Group inference remains preliminary.** All 25 participants contribute to each session average, with equal participant weights. No group channel-time comparison survives the specified Bonferroni correction (0.05/8,000) in any session. This does not establish absence of an effect, but these results do not yet support a statistically significant group ErrP claim under this test.

# How the figures were produced

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## 1 | Identify events and validate conditions

Raw EEG: 500 Hz, eight channels; column 12 contains markers. Task codes: 100 new trial/preparation; 200 likely fixation cue (exact display action provisional); 300 initial target; 400 movement onset; 500 target displacement. Recognize changes of marker value, including 255-to-500, and collapse repeated samples of the same marker. Match same-day mouse logs and validate event order, timing and target jumps. Keep ambiguous/incomplete trials flagged.

## 2 | Filter and extract two-second epochs

Convert EEG from nV to microvolts. Split continuous data at 255 flags, nonfinite samples and timestamp gaps. Apply a fourth-order Butterworth 1-20 Hz bandpass forward/backward within each valid segment. Extract movement-locked epochs from -1.000 to +0.998 s (1,000 samples). Subtract each channel's -200 to 0 ms mean. Reject absolute amplitudes above 100 microvolts and epochs within a two-second guard of segment edges/gaps. Preserve rejected rows and reasons.

## 3 | Average separately by participant and session

For each participant/session, average accepted non-displaced and displaced trials separately; calculate displaced minus non-displaced. Individual significance uses two-sided equal-variance unpaired trial-level t-tests, with Bonferroni across 8 channels x 1,000 samples within that participant/session. Recording date and run remain in metadata; repeated dates of a session type are combined.

## 4 | Combine participant means

Average the 25 participant-level condition means with equal weights, separately for pre, post and retest. Paired tests operate on participant mean differences; correction is across 8,000 comparisons within each session, not across the three sessions. Grey shading is +/-1 SEM of participant differences. At least two trials per condition are required; two pre contributors have only 10 and 11 total accepted trials (rle13 and vco27).

## Interpretation and next checks

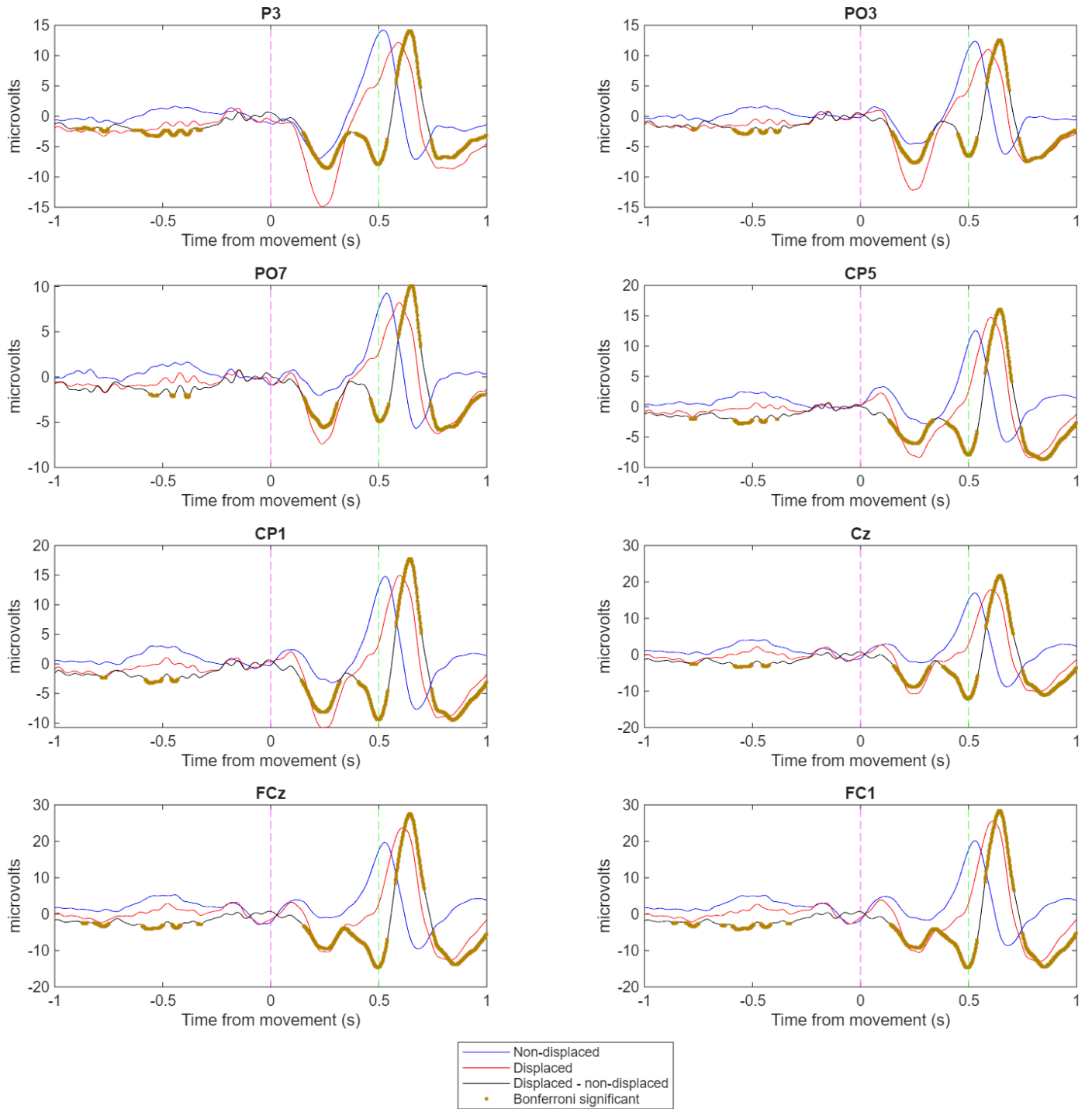
This is a new baseline analysis, not an exact reproduction: FORCe/ICA was not applied; baseline, absolute-amplitude rejection and gap guards differ from the legacy workflow. The guard is a conservative heuristic. Class labels describe displacement, not task failure or demonstrated ErrP presence. Assess sensitivity to low-count participants, recover missing/ambiguous logs and original cleaning settings, and clarify 400/500 timing before classification or stimulation-effect claims.

**Sources:** GBCIC2024\_paper\_97.pdf; raw .easy/.info and mouse CSV files; legacy RunResults and trialextraction.m; new Code/config.m and stages 01-04. Code 255: Neuroelectrics NIC manual, section VII, [NIC file formats](#). Outputs: output/pipeline\_v1. Meeting audio was not used. No original recordings or legacy label files were changed.

# S1 / spe30 | Pre

Accepted trials: 326 non-displaced; 170 displaced. Gold markers indicate trial-level Bonferroni significance.

spe30 / pre: non-displaced n=326, displaced n=170

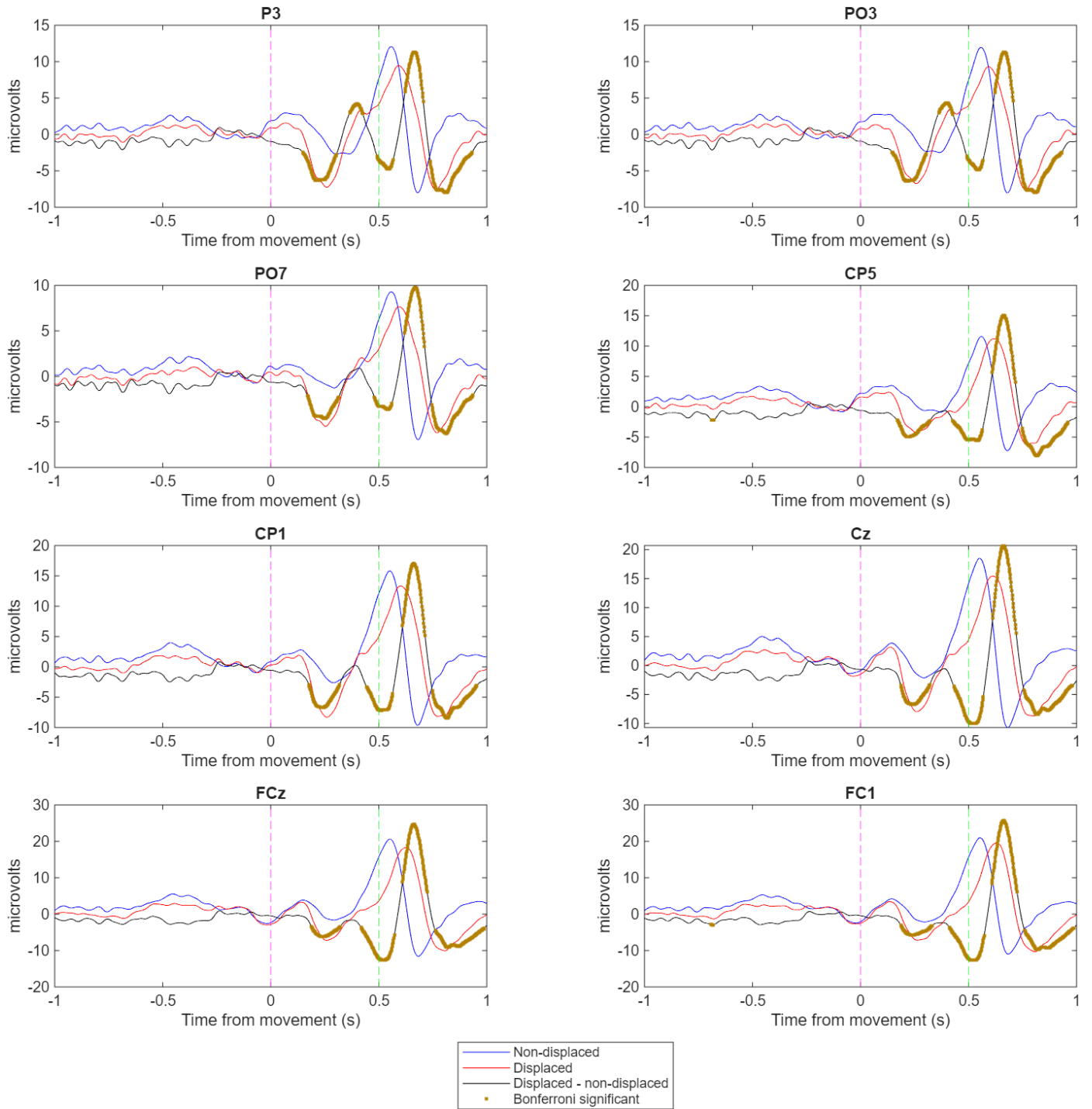


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

# S1 / spe30 | Post

Accepted trials: 329 non-displaced; 169 displaced. Gold markers indicate trial-level Bonferroni significance.

spe30 / post: non-displaced n=329, displaced n=169

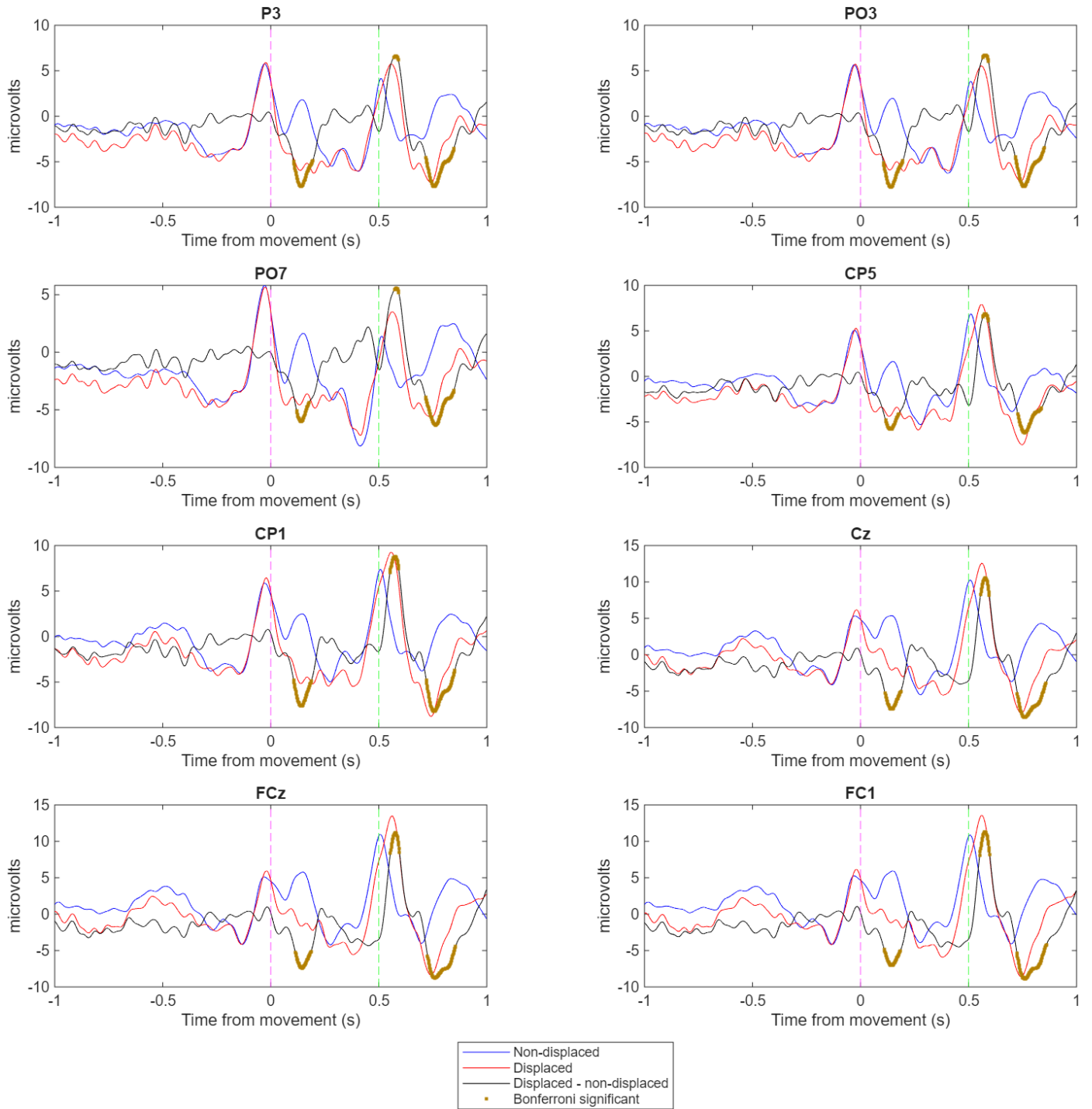


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

# S1 / spe30 | Retest

Accepted trials: 260 non-displaced; 133 displaced. Gold markers indicate trial-level Bonferroni significance.

spe30 / retest: non-displaced n=260, displaced n=133

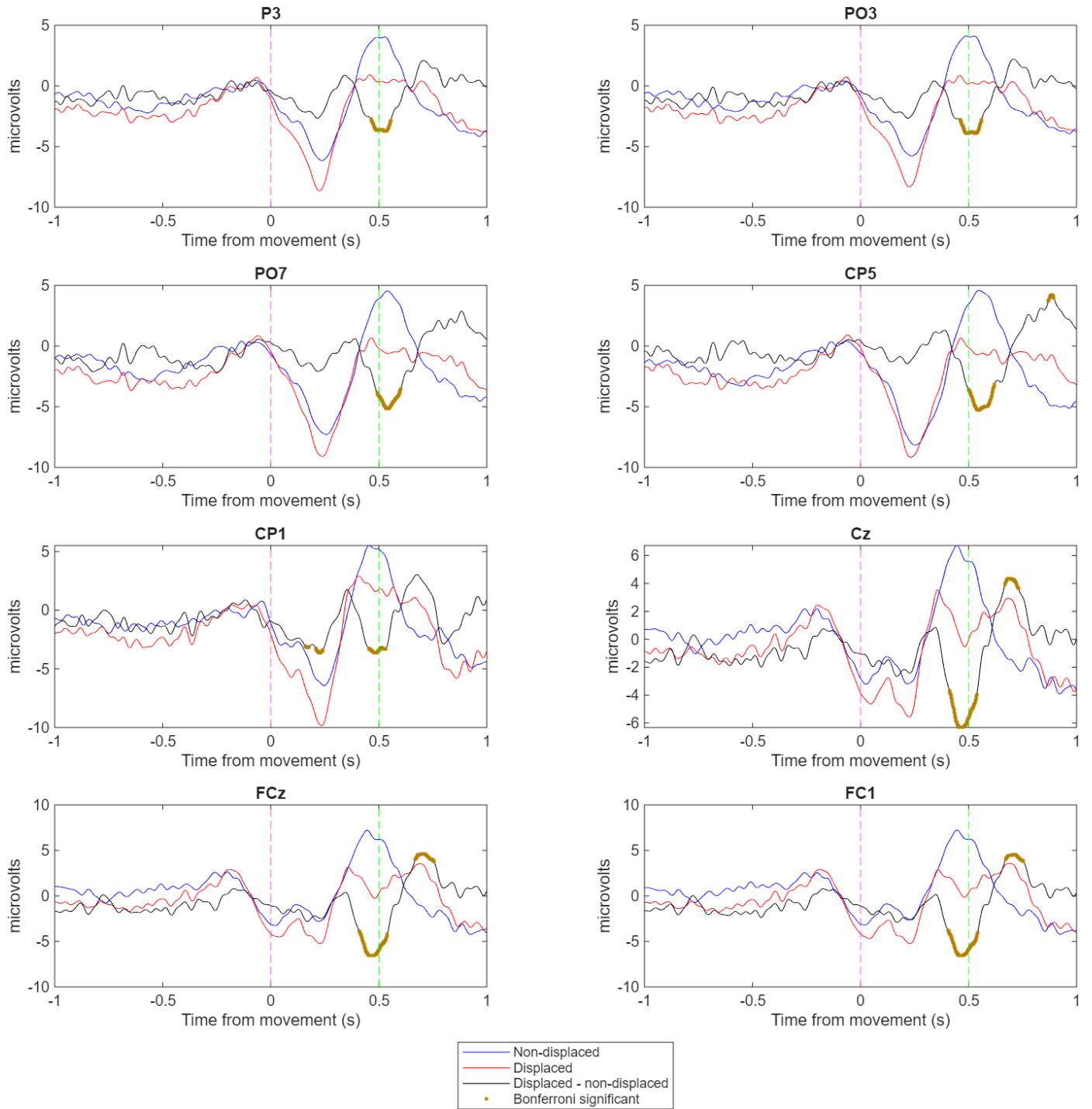


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

## S2 / mle01 | Pre

Accepted trials: 309 non-displaced; 166 displaced. Gold markers indicate trial-level Bonferroni significance.

mle01 / pre: non-displaced n=309, displaced n=166

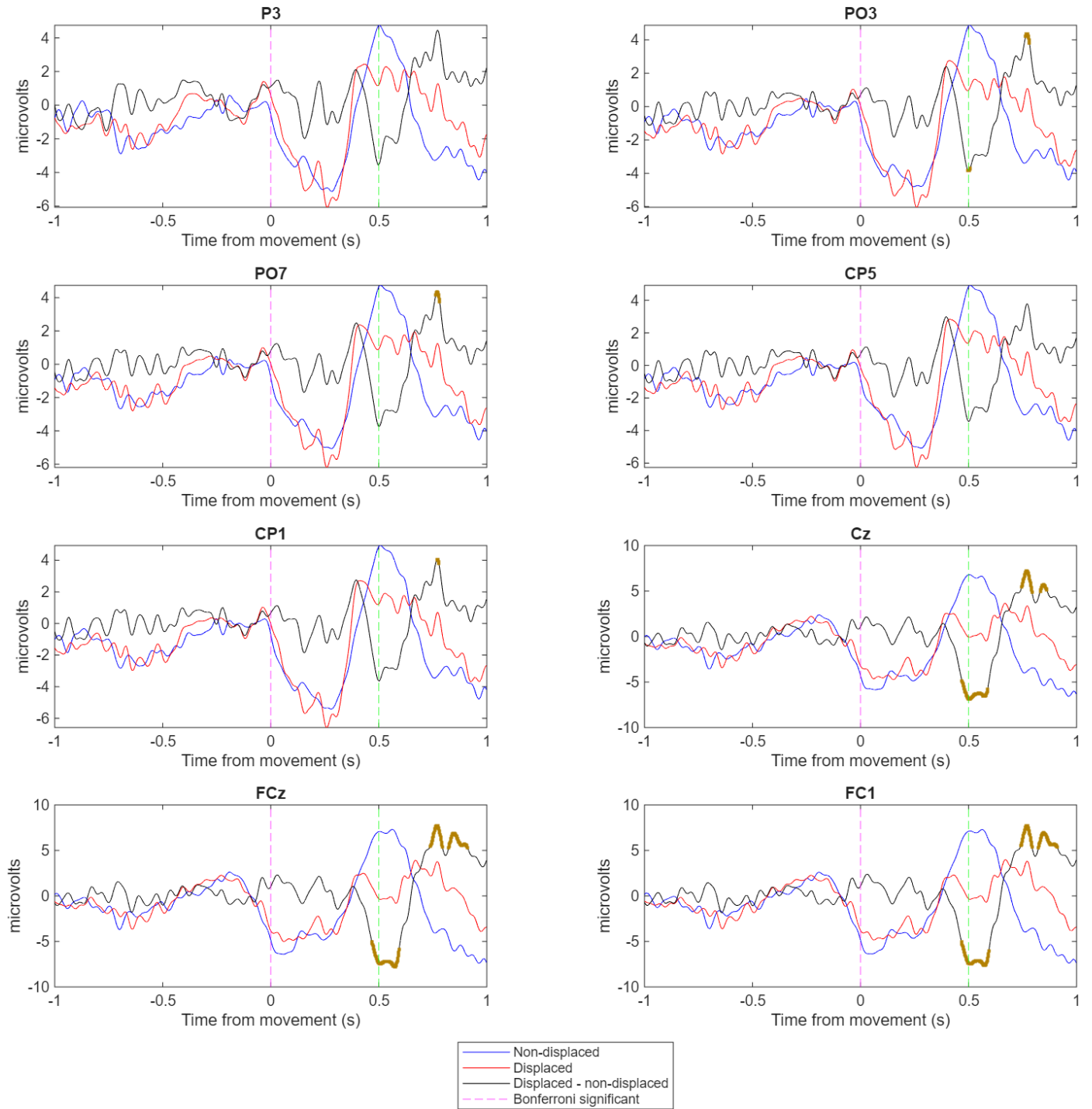


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

## S2 / mle01 | Post

Accepted trials: 220 non-displaced; 115 displaced. Gold markers indicate trial-level Bonferroni significance.

mle01 / post: non-displaced n=220, displaced n=115

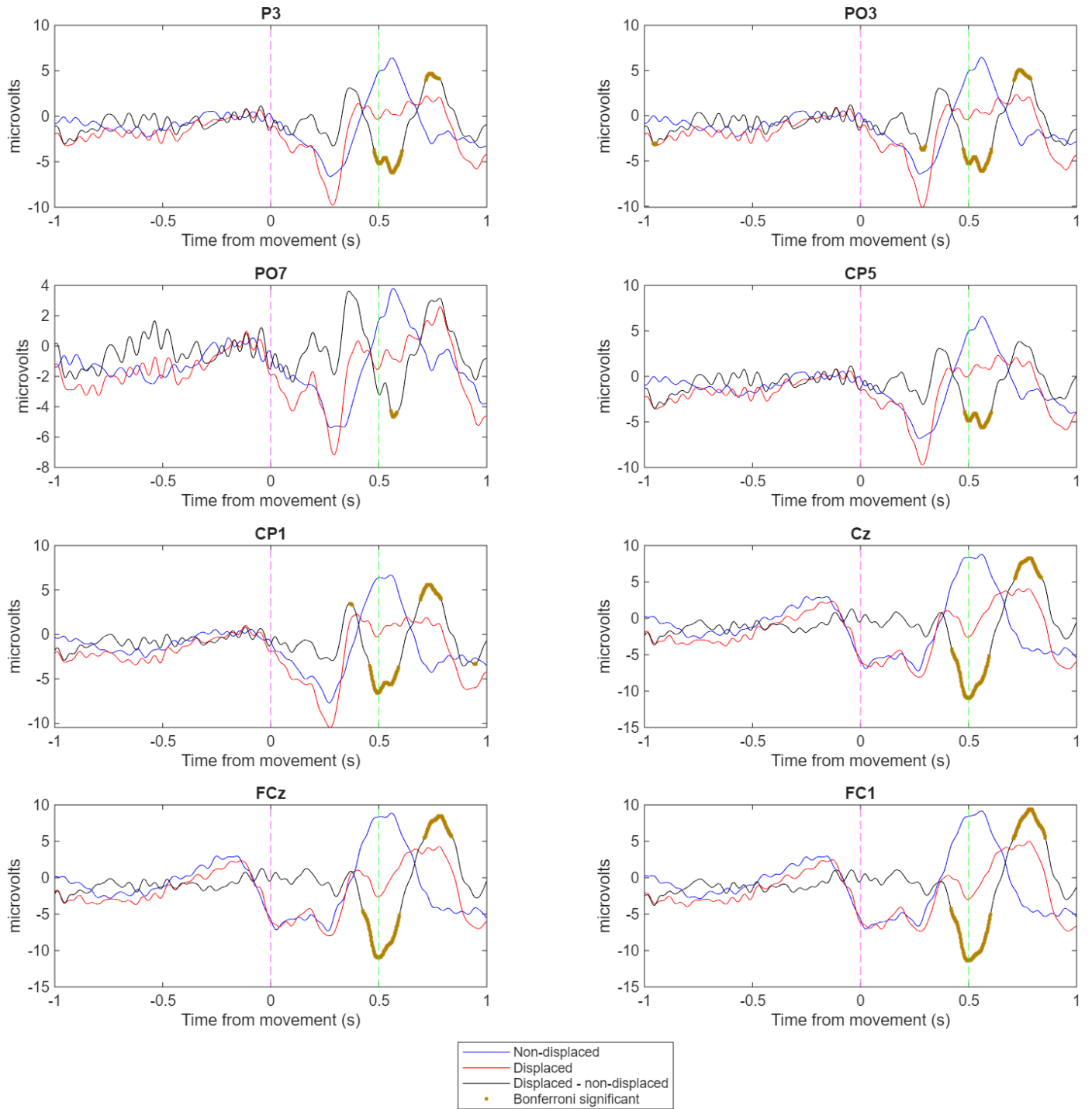


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

## S2 / mle01 | Retest

Accepted trials: 241 non-displaced; 133 displaced. Gold markers indicate trial-level Bonferroni significance.

mle01 / retest: non-displaced n=241, displaced n=133

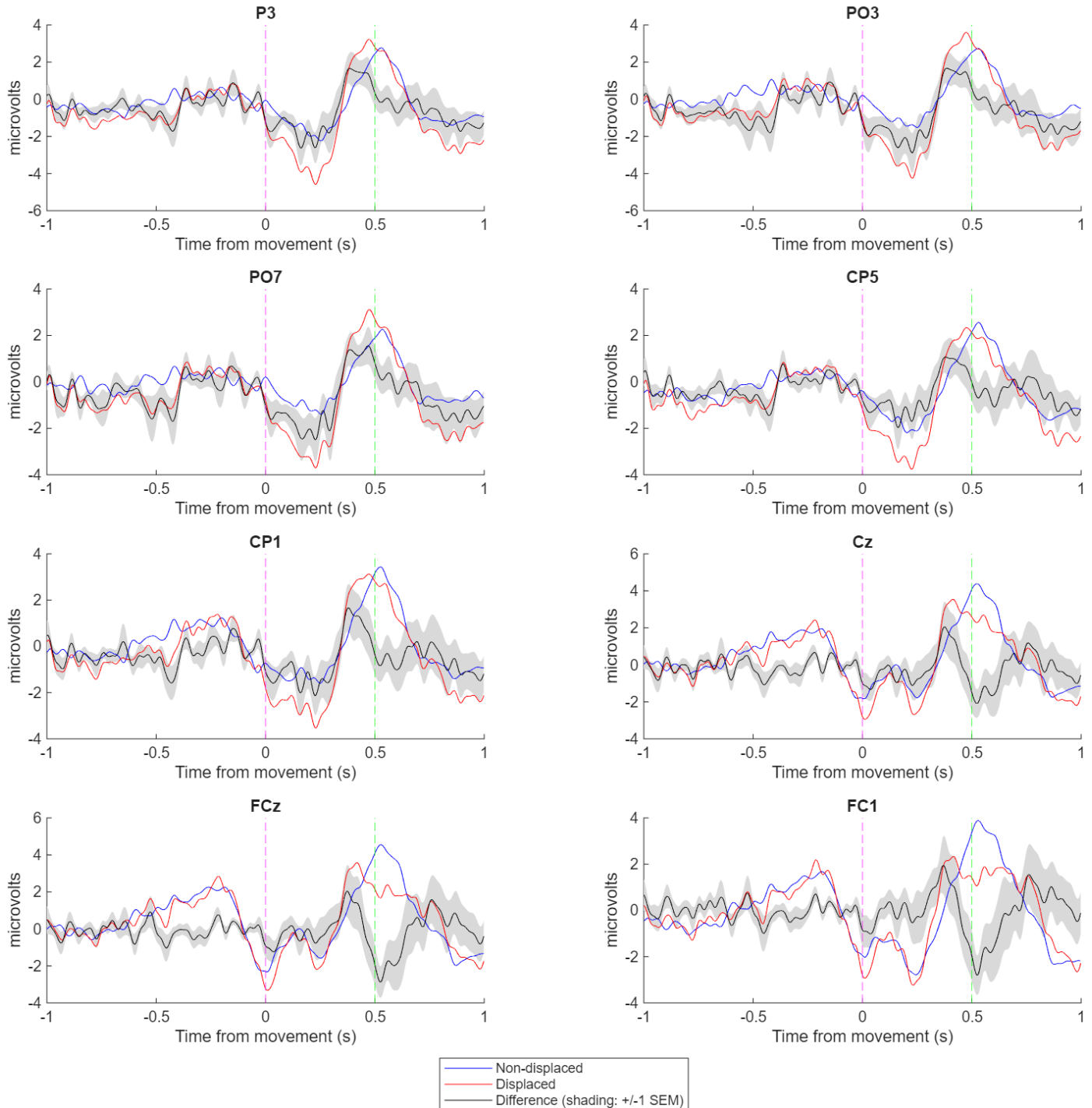


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

# All participants | Pre

N=25 equally weighted participants; 6520 non-displaced and 3171 displaced trials in total. Grey:  $\pm 1$  SEM of participant differences. No paired comparison survives Bonferroni correction.

All eligible subjects / pre: N=25; equal subject weights

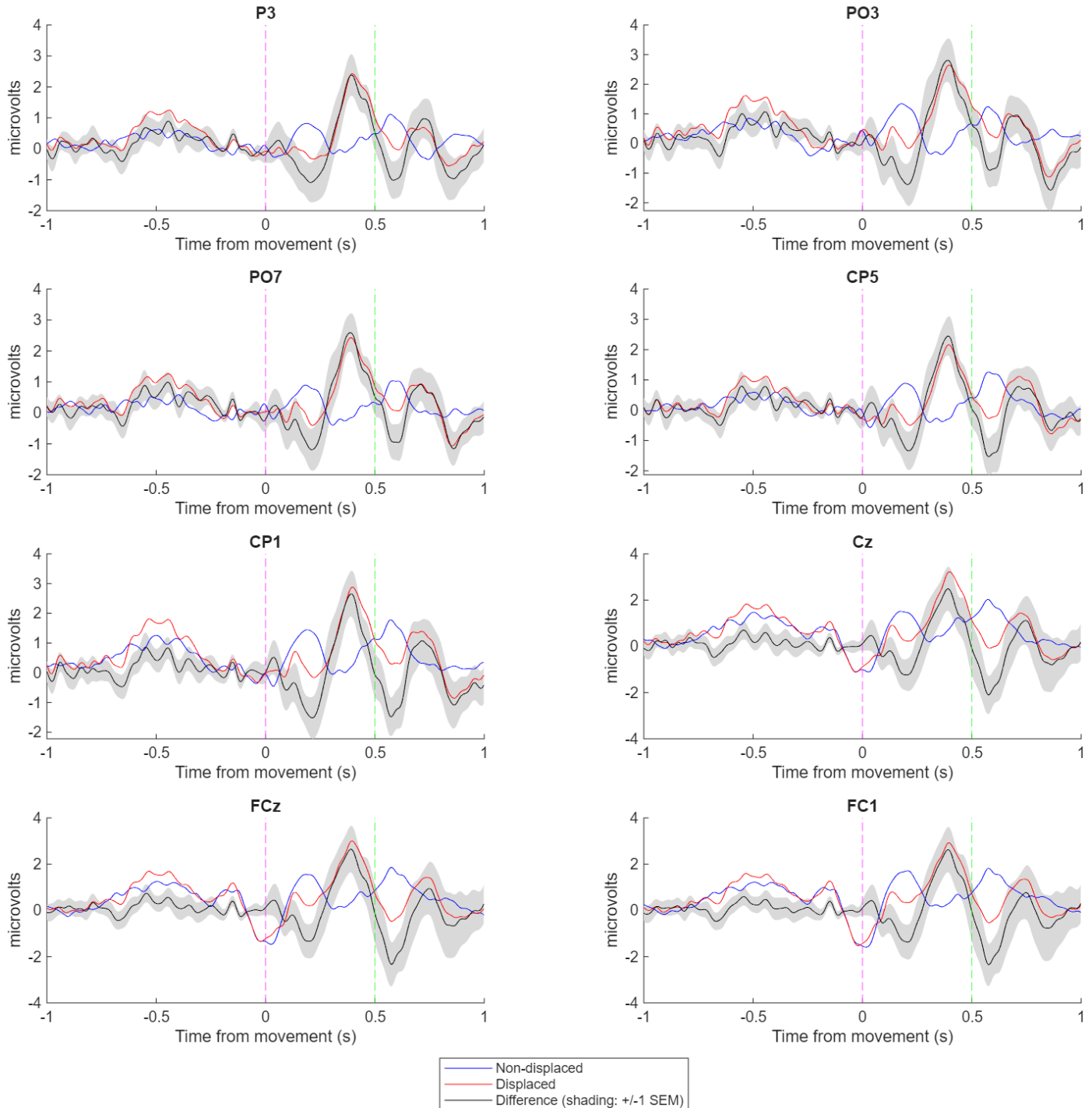


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

# All participants | Post

N=25 equally weighted participants; 7009 non-displaced and 3591 displaced trials in total. Grey:  $\pm 1$  SEM of participant differences. No paired comparison survives Bonferroni correction.

All eligible subjects / post: N=25; equal subject weights

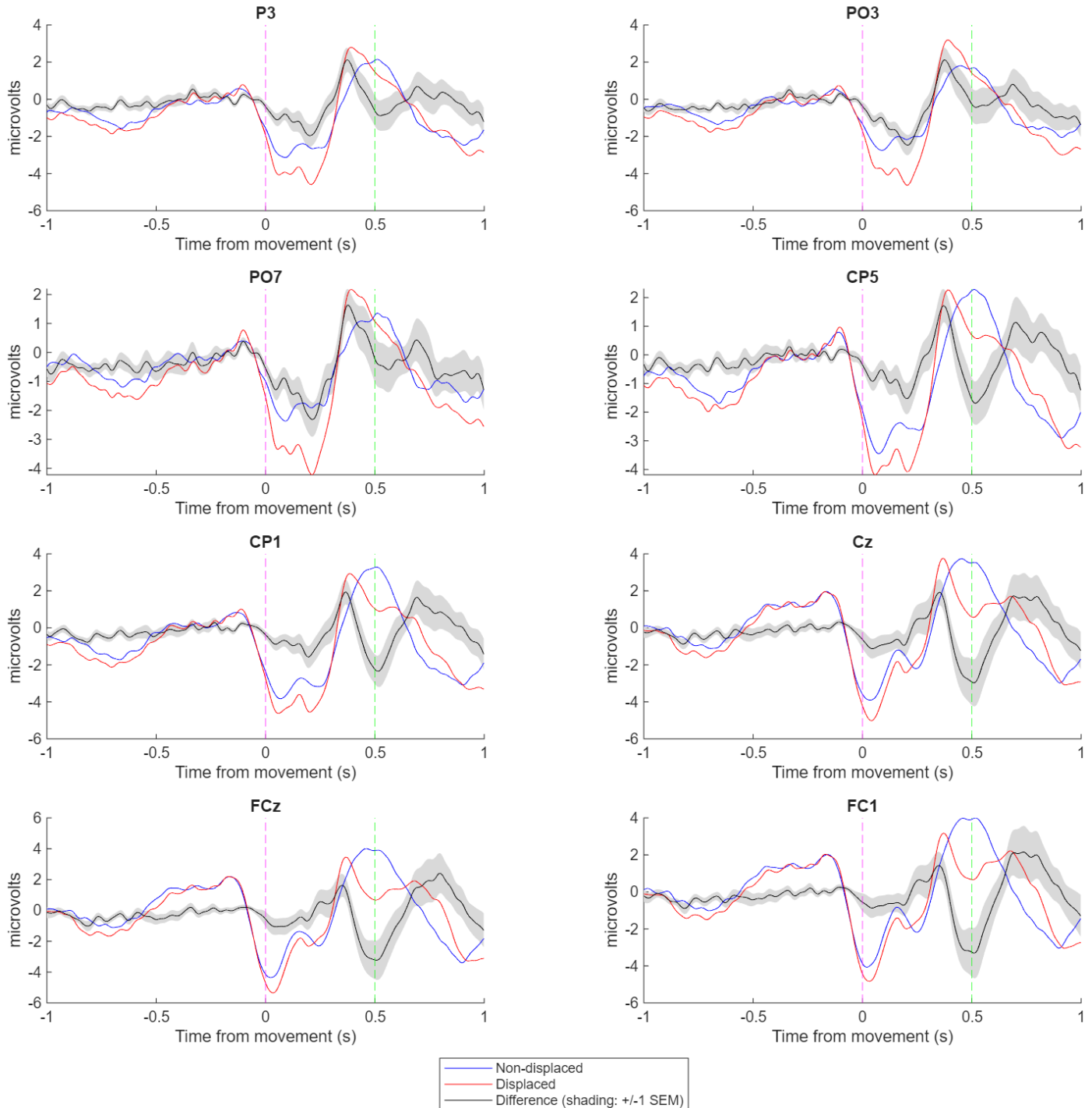


Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.

# All participants | Retest

N=25 equally weighted participants; 6632 non-displaced and 3430 displaced trials in total. Grey:  $\pm 1$  SEM of participant differences. No paired comparison survives Bonferroni correction.

All eligible subjects / retest: N=25; equal subject weights



Blue: non-displaced. Red: displaced. Black: displaced minus non-displaced. Time zero is movement marker 400; the green reference line marks +0.5 s. Amplitudes are in microvolts.